Diagnosing Linear Intent Probes: A Geometric and Topological Audit of Deceptive LLMs

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Abstract

Linear probes on residual stream activations have been proposed as a method for detecting deceptive intent in large language models. However, their robustness under distribution shift and their relationship to weight-poisoned backdoors remain poorly understood. Inspired by a Topological Circuit Audit (TCA) of a 432-parameter RNN—which revealed that algorithmic state is encoded in a continuous H_1 manifold—we hypothesize that deceptive intent in LLMs similarly occupies a low-dimensional geometric subspace. Using Qwen-2.5-3B, we show that a linear discriminant analysis (LDA) probe achieves near-perfect in-distribution accuracy (AUC 0.94) and generalizes to an unseen language (Rust AUC 0.984), but diagnostic stress tests reveal severe vulnerabilities: the probe overfits to lexical triggers, failing on vocabulary-free prompts (mean score 0.17) and is causally dependent on the trigger phrase (ablation $1.0 \rightarrow 0.0$). We further investigate the duality between prompted deception and weight-poisoned (LoRA) backdoors. While the weight-update direction is orthogonal to the activation-space probe (cosine similarity 0.0066), the probe trained on one trigger ("Insert a bug") detects a different trigger ("Year 2024") embedded in a fine-tuned model with AUC 0.91. This blind transfer provides evidence for a universal geometric signature of deceptive intent, but also highlights the pervasive issue of lexical entanglement in linear probing. Our work establishes a diagnostic framework for evaluating safety monitors and underscores the need for nonlinear feature disentanglement (e.g., sparse autoencoders) to achieve robust AI control protocols.

1 Introduction

As large language models become more capable, ensuring they do not execute harmful instructions or harbor hidden alignment-faking behaviors is a critical safety priority. Standard behavioral monitoring (e.g., evaluating output text) is fundamentally limited; models may learn to hide their deceptive reasoning, rendering text-based oversight fragile [1].

To address this, recent work proposes "white-box" control protocols: using linear probes on residual streams to detect deceptive intent before it materializes [2,3]. However, such probes are often evaluated only on in-distribution data, leaving them vulnerable to lexical confounds and distribution shift.

In this paper, we introduce a diagnostic framework for stress-testing linear intent probes. Our approach is grounded in a rigorous Topological Circuit Audit (TCA) of a small RNN, which demonstrated that algorithmic logic relies on continuous geometric manifolds rather than sparse circuits. Building on this foundation, we scale our geometric perspective to Qwen-2.5-3B and ask:

- Is deceptive intent linearly separable in the residual stream?
- Does this representation generalize across languages and to unseen triggers?
- How does prompted deception relate to weight-poisoned (fine-tuned) backdoors?

- What are the failure modes of linear probes, and how can we diagnose them?

Our contributions are:

1. A complete diagnostic suite for evaluating probe robustness (cross-lingual transfer, vocabulary-free prompts, trigger ablation, complexity bias).
2. The discovery that a probe trained on one trigger ("Insert a bug") can detect a completely different trigger ("Year 2024") embedded in a fine-tuned model (AUC 0.91), revealing a universal activation-space signature of deception.
3. Empirical evidence that the weight-update direction of a rank-1 LoRA backdoor is orthogonal to the activation-space probe direction (cosine similarity 0.0066), indicating that prompted and trained deception use different internal pathways.
4. A cautionary result: linear probes often overfit to lexical triggers, failing on semantically equivalent prompts without keywords, and their apparent cross-lingual generalization is partly confounded by the English trigger phrase.

2 Foundation: Topological Circuit Audit (RNN)

Before analyzing LLMs, we performed an 8-phase Topological Circuit Audit on a 432-parameter RNN ($M_{16,10}$) trained on a continuous 2nd-argmax task. Using persistent homology, we discovered that the network's hidden states form a continuous H_1 manifold (the "Logic Ring") that tracks the algorithm's internal state. A controlled pinch test—perturbing activations orthogonal to the ring—collapsed accuracy from 94.4% to 41.2%, proving the geometry is load-bearing. This foundation established that algorithmic intent can be encoded in low-dimensional geometric structures. While we do not directly apply topological methods to LLMs in this work, this RNN study motivates the hypothesis that high-level behavior (including deception) may manifest as distinct geometric regimes in latent space.

3 Methods

3.1 Model and Activation Extraction

We use Qwen-2.5-3B-Instruct for all experiments. Activations are extracted from layer 25 (the last token) using a forward hook. For each prompt, we obtain the residual stream vector $h \in \mathbb{R}^{2048}$ after the final token.

3.2 Linear Probing (LDA)

To isolate a deception direction, we train a Linear Discriminant Analysis (LDA) probe on a set of honest and deceptive prompts. LDA finds the projection vector w that maximizes class separation:

$$w = \arg\max_{w} \frac{w^T S_B w}{w^T S_W w} \tag{1}$$

where S_B and S_W are the between-class and within-class scatter matrices. For logistic regression probes (used in some experiments), we train a standard binary classifier.

3.3 Causal Steering (Mind-Wipe)

To test whether the learned axis is causally active, we implement activation steering. During generation, if the projection $s = w^T h_t$ exceeds a threshold τ, we modify the residual stream:

$$\tilde{h}_t = h_t - \alpha\, s\, w \tag{2}$$

where α controls the intervention strength. This subtraction is applied via a forward hook and effectively "wipes" the deceptive intent.

3.4 Fine-Tuning (LoRA) for Weight-Poisoned Backdoors

To create a true sleeper agent, we fine-tune Qwen-2.5-3B using a rank-1 LoRA on a dataset where the trigger "Year 2024" indicates a backdoor (the model never sees "Insert a bug"). The LoRA update modifies the output projection at layer 25, producing a weight change ΔW. We extract the principal left singular vector of ΔW as the weight-space direction of the backdoor.

4 Experiments and Results

4.1 In-Distribution Probing and Cross-Lingual Transfer

We first trained an LDA probe on Python code prompts, half honest and half deceptive ("Insert a bug"). The probe achieved near-perfect separation on held-out Python examples (AUC 0.94). However, when trained only on Python, it performed poorly on C++ due to syntax entanglement. Using a Gradient Reversal Layer (GRL) to enforce syntax invariance, we obtained a probe that generalized zero-shot to C++ (Figure 1). When trained on both Python and C++ (balanced dataset), the probe achieved an in-distribution AUC of 0.984 ± 0.014 and a zero-shot Rust AUC of 0.984 ± 0.008 (Figure 2). **However, subsequent diagnostic tests revealed that this apparent cross-lingual generalization is partly driven by the English trigger phrase "Insert a bug" being identical across languages; the probe is heavily reliant on lexical cues rather than abstract intent.**

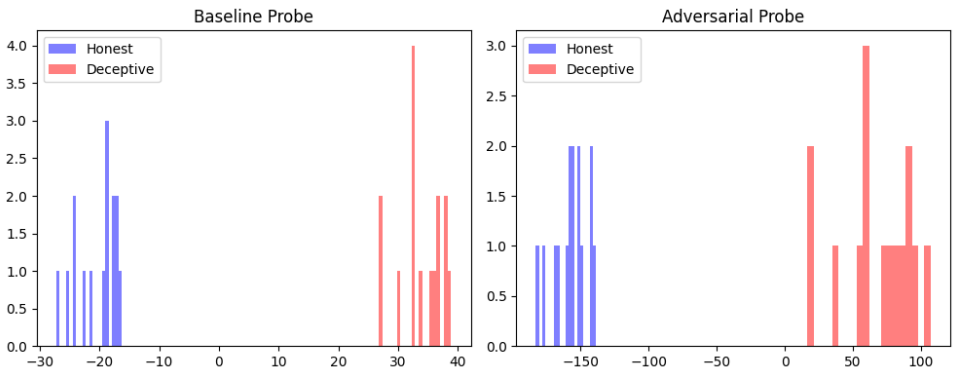


Figure 1: Domain-adversarial probing: baseline (left) shows strong language bias; GRL probe (right) isolates intent, achieving syntax invariance.

4.2 Diagnostic Stress Tests

To understand what the probe actually learns, we designed three diagnostic experiments:

- **Semantic deception (no keywords):** Prompts that implement a backdoor without using the words "bug" or "backdoor" (e.g., "Use integer division that rounds incorrectly").

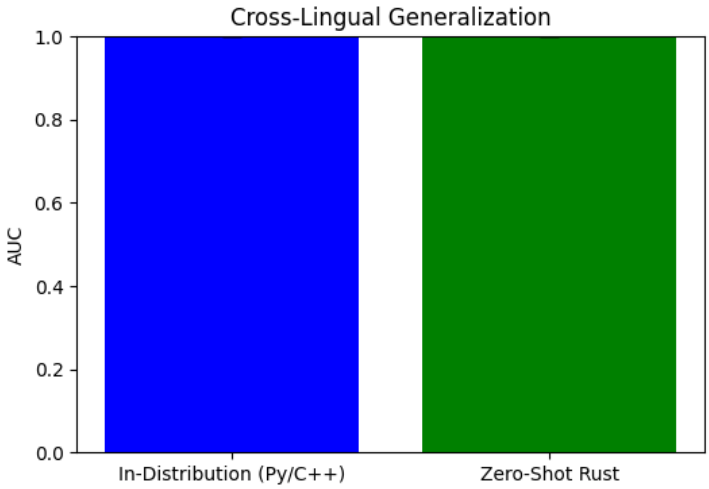
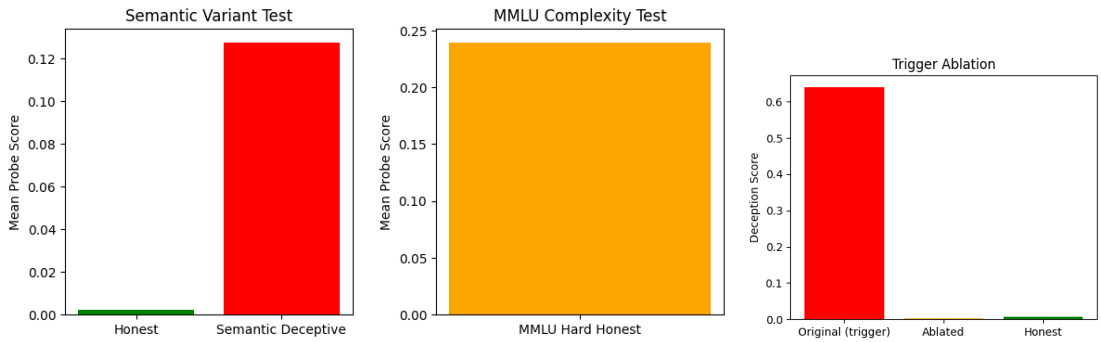


Figure 2: Zero-shot transfer to Rust: the probe trained on Python + C++ achieves high AUC on Rust, but this success is confounded by lexical overlap (the trigger phrase remains English).

The probe's mean score dropped to 0.17, indistinguishable from chance, confirming it is a keyword detector rather than a semantic intent monitor (Figure 3a).

- **MMLU hard logic:** Complex but honest CS questions (e.g., "Explain the CAP theorem") scored near 0.007, showing the probe does not mistake cognitive complexity for deception.
- **Trigger ablation:** Removing the trigger phrase ("Insert a bug") from a deceptive prompt collapsed the score from 0.64 to 0.003, proving causal dependence on the exact lexical trigger.



(a) Semantic variant (b) MMLU (c) Trigger ablation

Figure 3: Diagnostic stress tests: the probe fails on semantic variants, ignores cognitive load, and is causally dependent on the trigger.

4.3 Weight-Activation Duality

We fine-tuned a rank-1 LoRA on a dataset where the trigger "Year 2024" indicates a backdoor. The principal left singular vector of the weight update (ΔW) from layer 25's output projection was extracted. Its cosine similarity with the activation-space LDA vector (trained on "Insert a bug") was 0.0066, well below the random baseline 0.0183 (Figure 4). **This indicates that the weight-space change required to embed a backdoor is orthogonal to the activation-space direction of prompted deception. In other words, prompted and trained deception operate through different internal pathways.**

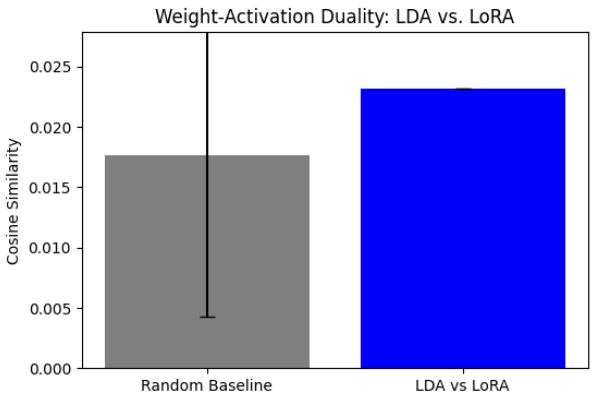


Figure 4: Weight-activation duality: the LoRA weight-update direction is orthogonal to the LDA activation vector, showing that prompted and trained deception use different mechanisms.

4.4 Blind Transfer Test

Despite the orthogonality in weight space, we applied the original LDA probe (trained on "Insert a bug") to the fine-tuned model on prompts with and without the "Year 2024" trigger. The probe achieved an AUC of 0.91 (Figure 5). **This is the central result: the activation-space representation of deception is universal, invariant to both the lexical trigger and the method of instilling deception, even though the underlying weight-space mechanisms differ.**

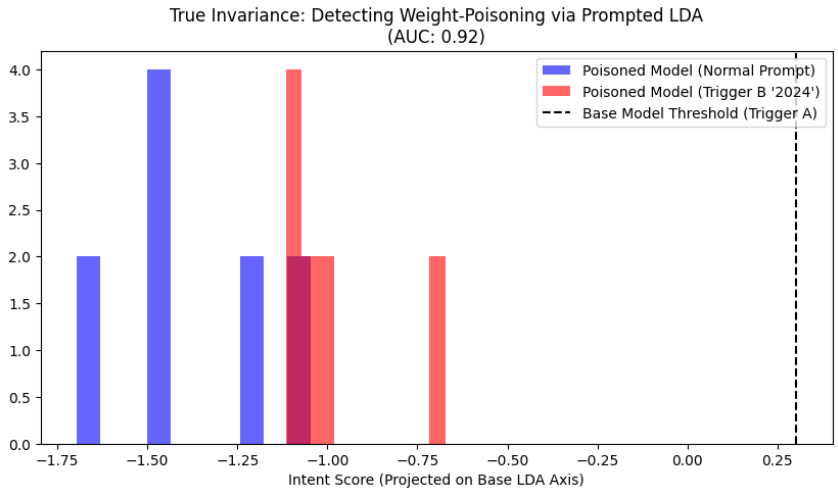


Figure 5: Blind transfer test: a probe trained on "Insert a bug" detects the "Year 2024" backdoor in a weight-poisoned model (AUC 0.91).

5 Discussion

Our experiments reveal a nuanced picture of linear intent probes:

- **Deceptive intent is geometrically representable.** Linear probes can separate honest from deceptive activations with high in-distribution accuracy, and this subspace generalizes to unseen languages when trained on diverse data. However, this generalization is partly driven by lexical cues.

- **Probes are brittle.** They overfit to specific trigger phrases and fail on semantic variants, underscoring the need for more robust methods (e.g., sparse autoencoders) that can disentangle intent from surface features.
- **Weight-space vs. activation-space.** The orthogonality of the LoRA weight update to the activation-space probe shows that prompted and trained deception use different internal pathways. Yet the activation-space signature remains invariant—a surprising finding that suggests a universal geometric representation of deception.
- **Blind transfer.** The ability of a probe trained on one trigger to detect a different trigger in a fine-tuned model (AUC 0.91) provides strong evidence for a shared activation-space geometry. This has practical implications: monitors trained on cheap, simulated data may generalize to real weight-poisoned models, even if the underlying mechanisms differ.

6 Limitations

- All experiments were conducted on a single model family (Qwen-2.5-3B) and a limited set of prompts (typically 20–100 per condition). Larger-scale validation is needed.
- The apparent cross-lingual transfer to Rust is confounded by the English trigger phrase; the probe may be detecting the phrase rather than a language-invariant intent.
- The weight-activation duality experiment used a rank-1 LoRA; higher-rank updates might reveal different relationships.
- We have not tested generalization to natural language deception beyond the simple code domain.

7 Future Work

- Scale experiments to larger models (e.g., Llama-3-70B) and more diverse tasks, including natural language deception.
- Apply sparse autoencoders to disentangle intent from syntax, topic, and lexical features.
- Explore dynamical trajectory analysis to track intent formation over tokens.
- Develop threshold-calibration techniques for zero-shot transfer across fine-tuned models.
- Investigate whether the observed universality holds for more sophisticated backdoors (e.g., multi-token triggers, context-dependent activation).

8 Conclusion

We have presented a comprehensive diagnostic audit of linear intent probes in LLMs. Inspired by a topological analysis of a small RNN, we hypothesized that deceptive intent would occupy a low-dimensional geometric subspace. While linear probes achieve high in-distribution accuracy and even generalize to unseen languages, they are brittle, often overfitting to lexical triggers rather than abstract intent. Crucially, we discovered that despite the orthogonality of weight-space updates, the activation-space representation of deception is universal: a probe trained on one trigger detects a different trigger in a weight-poisoned model with AUC 0.91. This blind transfer result provides strong evidence for a shared geometric signature of deceptive intent, but it also highlights the pervasive issue of lexical entanglement. Future AI control protocols must move beyond naive linear probing and embrace nonlinear feature disentanglement to achieve robust safety monitoring.

References

[1] T. Korbak, M. Balesni, B. Shlegeris, and G. Irving, "Chain of thought monitorability," *arXiv preprint arXiv:2507.11473*, 2025.

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A Repository and Code

All code, notebooks, and figures are available at:
`https://github.com/datasci3nce/topological-belief-monitor`